Can Gravitational Waves be a Probe for Quantum Spacetime

Kumar S. Gupta

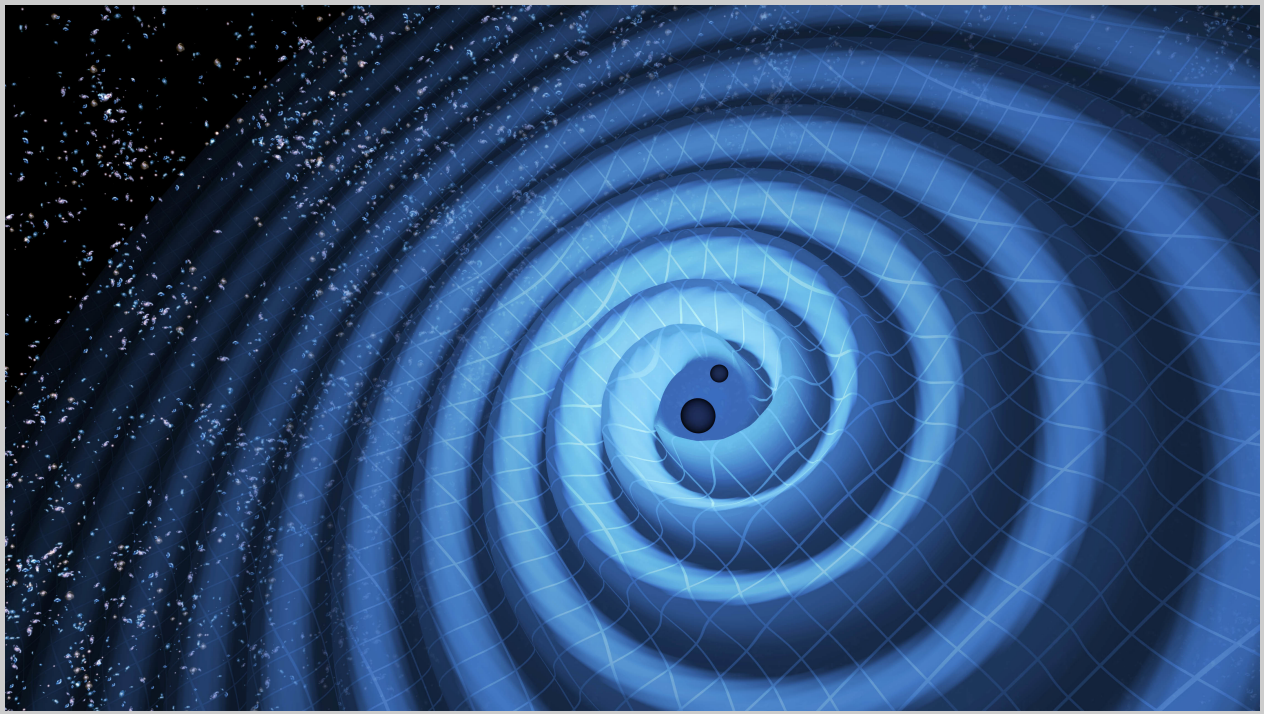
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Overview of the talk

- Gravitational waves
- Black hole ringdown and Quasinormal Modes
- Quantum spacetime
- Quasinormal modes as probes of quantum spacetime
- Supermassive black holes
- Outlook

Gravitational waves

- Gravitational wave (GW) was first predicted by Einstein in 1916.
- According to General Relativity, GW are ripples in space-time caused by massive accelerating objects.



Gravitational waves

- These transverse waves propagate with a speed of light and carry information about the sources that produce them.
- As the gravitational coupling G is very weak, the amplitudes of gravitational waves are very small.
- Hence gravitational waves produced by very massive astrophysical objects have better chance of being detected on earth.

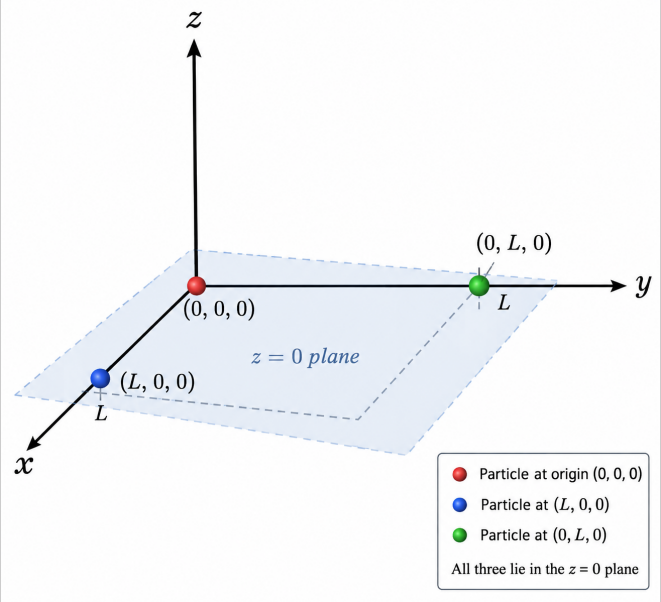
Gravitational waves

- Another feature of GW is that they are quadrupolar in nature. This is due to the energy-momentum conservation in closed systems.
- Eddington in 1922 obtained the quadrupole formula for the GW with all correct factors and for the first time applied the formalism to the mechanical system of a rotating bar. The time dependent GW wave is given by

$$h(t) = \frac{2G}{c^4}\frac{1}{r}\frac{\partial^2 Q}{\partial t^2}$$

  Q is the quadrupole moment of the source mass distribution and r is the distance between source to the detector. Formula valid for non-relativistic velocities.
- In mid 1960's Kip Thorne and John Wheeler started to apply the theory of GW to astrophysical systems.

Metric purturbation - wave equation

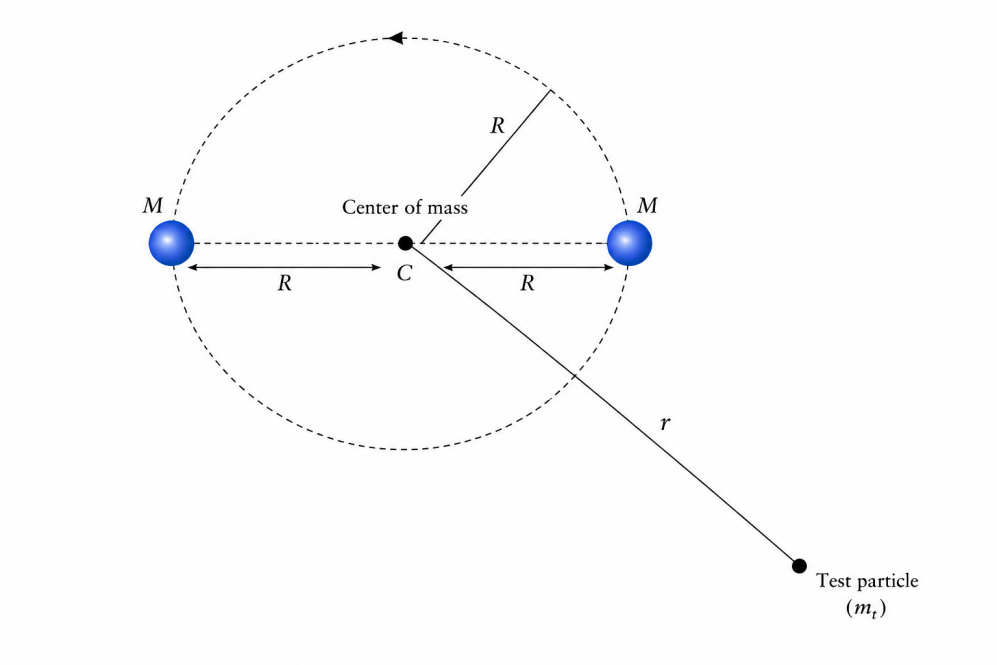


Perturbed metric $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ $\Box h_{\mu\nu} = 0$
Only two independent components of $h_{\mu\nu}$ satisfy

$$h_{xx} = -h_{yy} = h \sin \omega (t - \frac{z}{c})$$

As the wave passes along the z direction, the proper distance between the particles vary with time. The relative change in length along x axis : $\frac{\Delta L}{L} = \frac{1}{2} h \sin \omega t$ and along y axis : $\frac{\Delta L}{L} = -\frac{1}{2} h \sin \omega t$

Metric purturbation - order of magnitude



Consider two compact objects of mass M in a circular orbit of radius R. Amplitude of the gravitational wave is given by $h = \frac{2GM}{c^2R} \times \frac{2GM}{c^2r}$. For a solar mass object, $R_{sch} = \frac{2GM}{c^2} \sim 3$km and considering $R \sim 300$ km and $r \sim 1$ kPC $\approx 3X10^{19}$ metres, the amplitude of the metric perturbation is as small as $h \sim 10^{-22}$.

Gravitational waves - first evidence

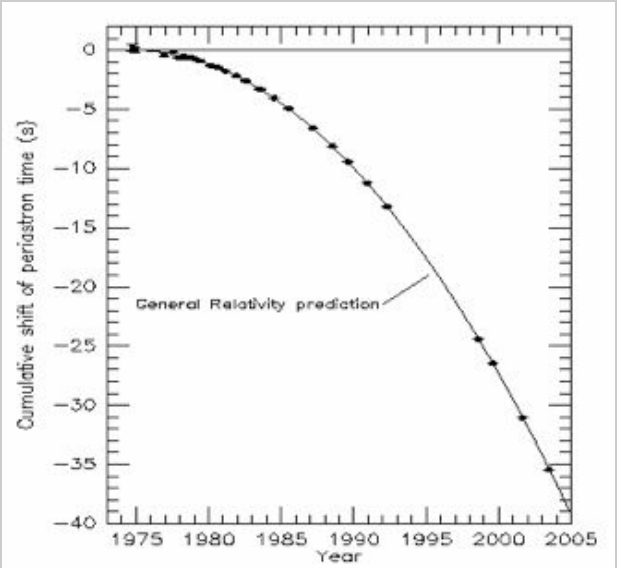
The first indirect evidence of GW came from pulsars, which are rapidly rotating highly magnetic neutron stars.

This movie shows dynamic rings and jets of matter around the pulsar in the Crab Nebula as observed in X-ray light by Chandra (left, blue) and optical light by Hubble (right, red).

Ref: science.nasa.gov

Gravitational waves - first evidence

- In 1974, Russel Hulse and Joseph Taylor discovered a pulsar in a compact binary system and by studying that system over the years they showed that the orbital radius between the binary components was shrinking in agreement with the energy loss via gravitational waves as predicted by GR.



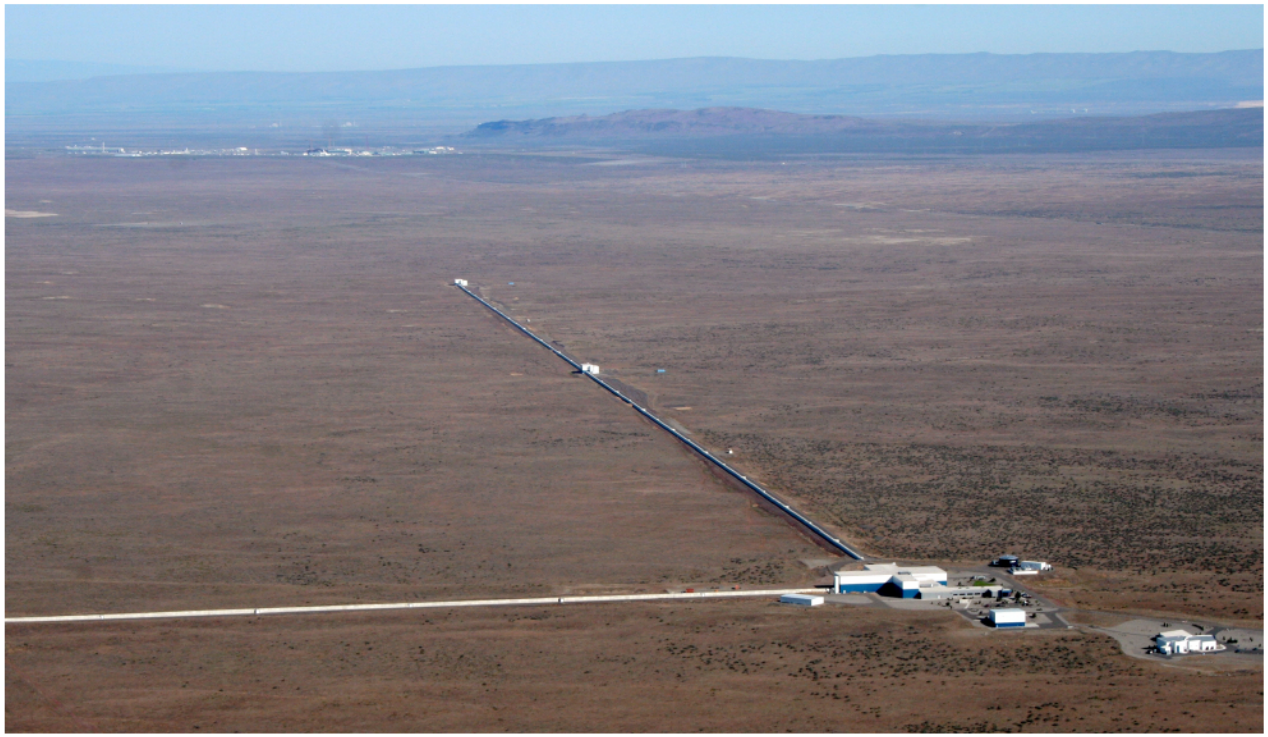
Ref: Weisberg and Taylor, arXiv:astro-ph/0407149

- This was the first (indirect) evidence of GW for which they received Nobel prize in 1993.

Gravitational waves - LIGO

- In 1975, Kip Thorne and Rainer Weiss developed the idea of using a laser based Michelson interferometer as a tool to detect GW, which later went and became what is now known as LIGO.
- In 1994, under the leadership of Barry Barish, the construction of the two LIGO obervatories in Livingston (Louisiana) and in Hanford (Washington) were started.
- On 14 September, 2015 the LIGO detectors in both Livingstone and Hanford detected GW from a merger of two black holes approximately of 30 solar masses each, an event named **GW150914** that happend at a distance of more than a billion light-years from earth.
- For this discovery, Barish, Thorne and Weiss were awarded the Nobel prize in 2017.

Hanford Observatory



Gravitational waves - LIGO Interferometer

- A laser beam of light is generated and directed toward a beam splitter, which splits it into two separate and equal beams. The light beams then travel perpendicularly to a distant mirror, with each arm of the device being 4 kilometers in length. The mirrors reflect the light back to the beam splitter, repeating this process about 200 times.
- When no gravitational waves are present, the distance between the beam splitter and the mirror is precisely controlled so that the light waves are kept out of phase with each other and cancel each other out
- When gravitational waves pass through the system, the distance between the end mirrors and the beam splitter lengthen in one arm and at the same time shorten in the other arm in such a way that the light waves from the two arms go in and out of phase with each other. This causes the signal to flicker.

LIGO Interferometer - Video

LIGO - a Precision Ruler

- Each arm of the Ligo interferometer is 4 km long.
- LIGO can detect a change in length of the arm upto about $\frac{1}{1000}$th of the width of a proton, which is about 10^{-18} metres.
- As the sensitivity increasese, LIGO will be able to detect a change in distance of upto $\frac{1}{10000}$th of the width of a proton.
- This is equivalent to detecting a change in distance to the nearest star (about 4.2 light years away) by the width of a human hair.
- This is the level of precision that is necessary it to measure the very weak gravitational waves.
- LIGO thus provides the most precise ruler ever created.

What event is detected by LIGO

- On 15 September, 2014, Ligo detected the GW from two merging black holes, with one slightly larger than the other. The event took place 1.3 billion years ago. This event is called GW150914.
- LIGO's instruments are designed to detect a specified range of frequencies of gravitational waves. In a binary inspiraling system of black holes, the wave radiated slowly increases in frequency as the objects move closer together. At some point, the pair will be orbiting each other at a rate that LIGO can finally detect, and the radiating gravitational waves will enter the sensitivity range.
- It turns out that by the time this happens, the objects are just seconds away from colliding. Consequently, the time they spend orbiting and emitting gravitational waves in LIGO's detectable range is typically very brief, ranging from a fraction of a second to tens of seconds.

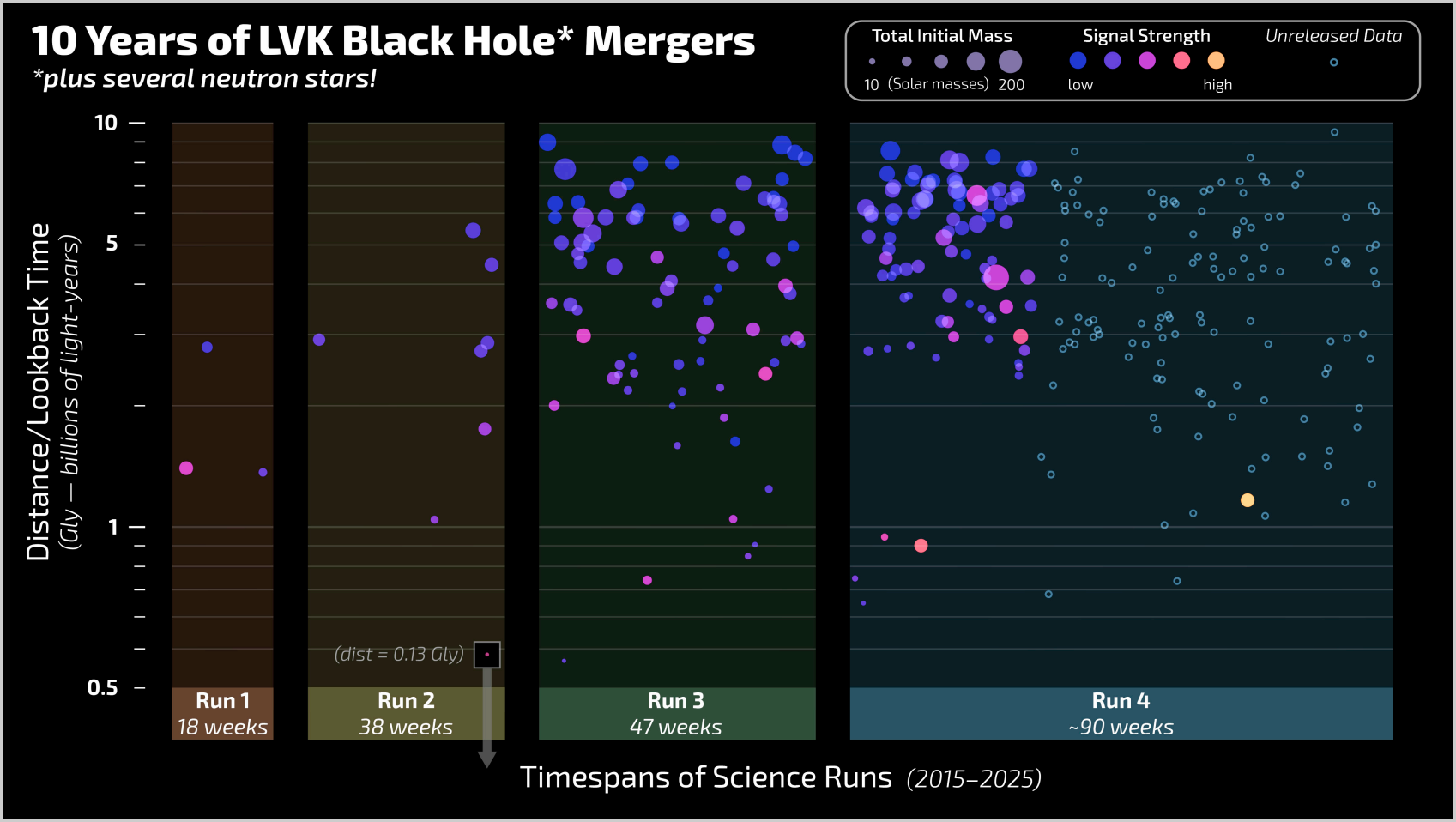
BH Merger Simulation

What event is detected by LIGO

- Based on the observed signals, it was estimated that the black holes were about 29 and 36 times the mass of the sun, and the event took place 1.3 billion years ago.
- About 3 times the mass of the sun was converted into gravitational waves in a fraction of a second—with a peak power output about 50 times that of the whole visible universe.
- By looking at the time of arrival of the signals—the detector in Livingston recorded the event 7 milliseconds before the detector in Hanford—scientists can say that the source was located in the Southern Hemisphere.

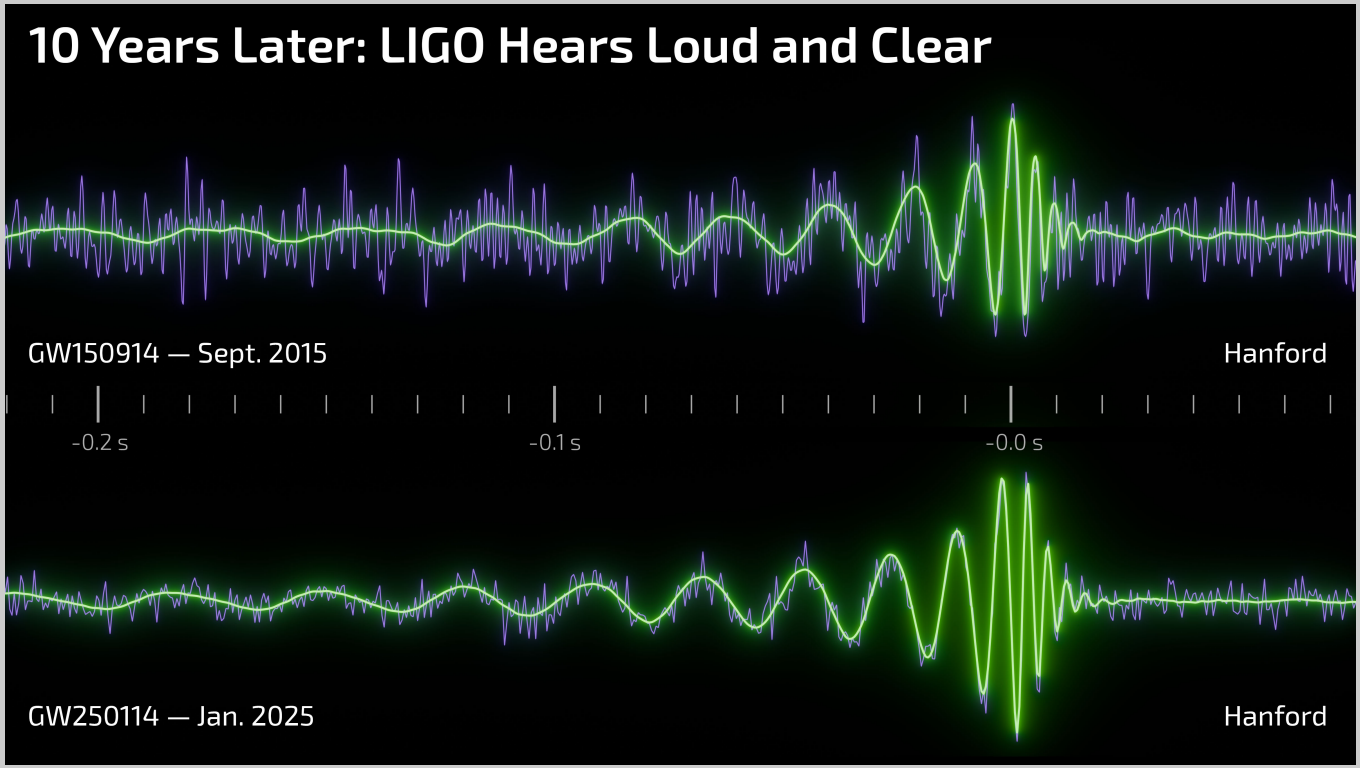
Audio Mapping

BH merger events 2015-2025



Ref: https://www.ligo.caltech.edu

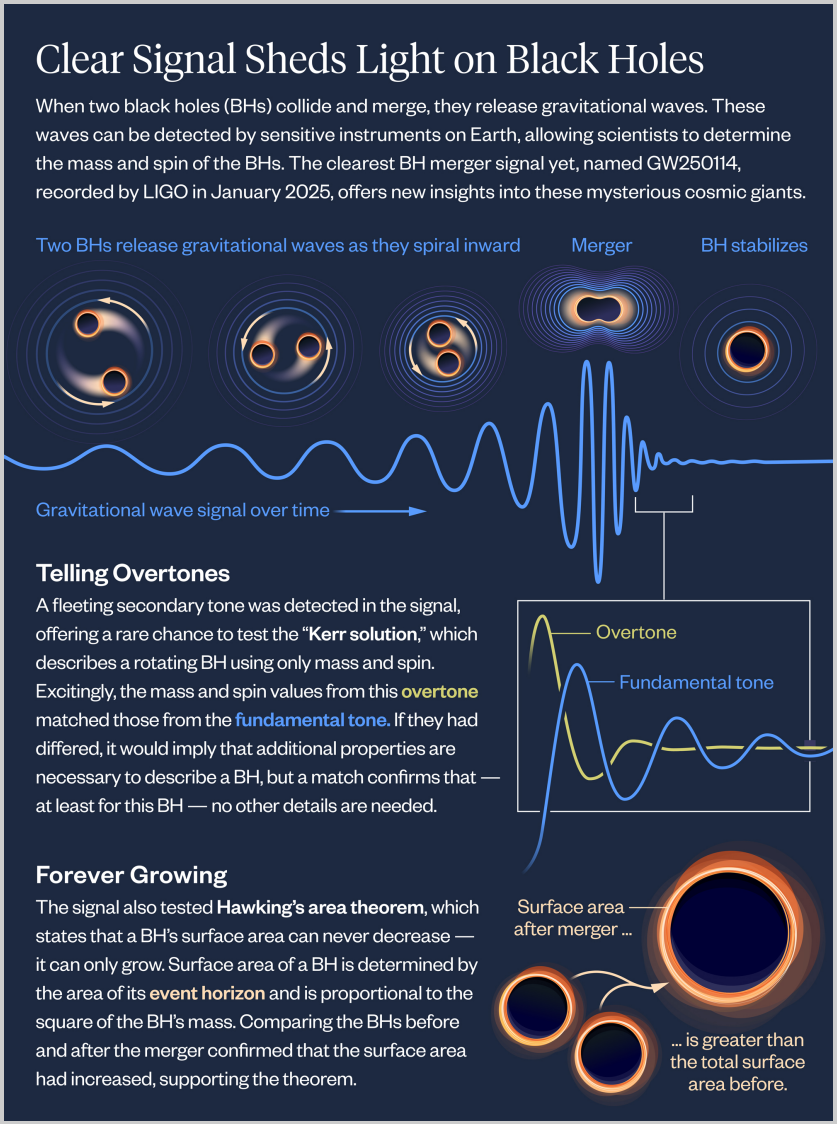
LIGO-Noise Reduction



This plot shows gravitational-wave signals recorded by the LIGO Hanford detector almost ten years apart. The top shows data from LIGO's first-ever detection of gravitational waves, an event called GW150914, captured in 2015. The bottom shows the signal known as GW250114, captured in 2025. Both events involve colliding black holes about 1.3 billion light-years away. The purple line shows the data, which are a combination of the signal plus background detector noise. The noise comes from a variety of sources, including seismic motions that jiggle giant mirrors inside LIGO. The green line shows the best-fit prediction from general relativity for each signal. The much lower noise seen today is thanks to cutting-edge improvements made to the LIGO detectors that hush unwanted noise.

Ref: https://www.ligo.caltech.edu

LIGO-GW250114



Ref: https://www.ligo.caltech.edu

GW in the Ringdown phase - Quasinormal modes

- According to GR, the newly formed black hole settles down to its final stable state by releasing sinusoidally damped gravitational waves, called quasinormal modes (QNM), with a fundamental mode and several overtones.
- These have actually been observed from the LIGO data for example for the initial event GW150914 as well as for later ones such as GW190521 and many more.
- With multiple modes arising from the same event, it is possible to verify that they depend only on the mass and spin of the black hole, which validates the no hair theorem.
- We shall henceforth focus on the QNM in the ringdown phase, which will be used to explore the quantum spacetime

GW in the Ringdown phase - Quasinormal modes

- QNMs are solutions of classical perturbation equations in a given gravitational background subject to specific boundary conditions. For asymptotically flat spacetime the solution is purely ingoing at the horizon and purely outgoing at spatial infinity.
- The QNM's have a real part representing the frequency and an imaginary part representing the damping. They reprsent time evolution of perturbations which are dominated by damped oscillations.
  $\omega = \omega_R + i\omega_I, \quad \psi(t) \sim e^{-i\omega_R t} e^{\omega_I t}, \quad \omega_I < 0$ implies stability.
- Frequency and damping depend on BH parameters and do not depend on the details of the initial perturbation.

QNM -Schwarzschild BH

The metric for an asymptotically flat, spherically symmetric and static black hole in 3+1 dimensions is given by

$$ds^2 = -f(r)dt^2 + f^{-1}(r)dr^2 + r^2 d\Omega^2 = g_{s\,ij} dx^i dx^j$$

For Schwarzschild black hole of mass M, $f(r) = \left(1 - \frac{2M}{r}\right)$.
In order to find gravitational perturbations for the Schwarzschild metric one needs to solve the linearized vacuum Einstein equation

$$R_{\mu\nu}(g) = 0$$

where $R_{\mu\nu}$ is the Ricci tensor that depends on the metric $g = g_s + h$ and g_s is the Schwarzschild background metric, while h is the metric perturbation. We assume a decomposition into tensor spherical harmonics for each $h_{\mu\nu}$ of the form

$$\psi(t, r, \theta, \phi) = \sum_{lm} \frac{\psi_{lm}(r, t)}{r} Y_{lm}(\theta, \phi)$$

QNM - Schwarzschild BH

With this, the wave equation for the QNM reduces to

$$\frac{d^2\psi}{dr_*^2} + \Big(\omega^2 - V(r)\Big)\psi = 0$$

where $V(r)$ is the Regge-Wheeler potential given by

$$V(r) = \frac{(r-R)\big(\ell(\ell+1)r - 3R\big)}{r^4}, \quad R = 2M$$

and the tortoise coordinate r_* is given by

$$r_* = r + 2M\log(\frac{r}{2M} - 1)$$

The wave equation is solved with the QNM boundary conditions : ψ is purely ingoing $r_* \to -\infty$ (horizon) and purely outgoing at $r_* \to \infty$ (spatial infinity).

QNM

- The QNM equation for Schwarzschild BH cannot be solved analytically, but various numerical procedures exist.
- The wave equation can also be solved within the WKB approximation.
- Typically it is found that real part of the QNM frequency is positive while the imaginary part is negative.
- This menas that in 3+1 dimensions, the Schwarzschild BH is stable against gravitational perturbations.
- We shall subsequently study the QNM's as a probe of the quantum space-time.

Quantum Space-Time

- Two basic questions about nature are:
  - Can space-time events be measured with arbitrary precision ?
  - Is there a fundamental and elementary length scale in nature ?
- These issues are related to the quantum structure of space-time relevant at the Planck scale.
- There are various candidates to describe such a regime, such as String Theory, Loop Quantum Gravity, Noncommutative Geometry and so on.
- All these theories strive to incorporate gravity and quantum theory within a unified and consistent framework.

Space-time UR

Heisenberg's Principle

$+$ $\Longrightarrow$ Space-time uncertainty relations

Einstein's Theory

- Measuring a space-time event with an accuracy δ causes and uncertainty in the momentum $\sim \frac{1}{\delta}$.
- Neglecting rest mass, an energy of the order $\frac{1}{\delta}$ is transmitted to the system and concentrated for some time in the localization region. The associated energy-momentum tensor generates a gravitational field.
- The smaller the uncertainties in the measurement of coordinates, the stronger will be the gravitational field generated by the measurement.

Space-time UR

- To probe physics at Planck Scale l_p, the Compton wavelength $\frac{1}{M}$ of the probe must be less than l_p, hence $M > \frac{1}{l_p}$, i.e. Planck mass.
- When this field becomes so strong as to prevent light or other signals from leaving the region in question, an operational meaning can no longer be attached to the localization.
- Similarly, observations of very short time scales also require very high energies. Such observations can also limit spatial resolutions leading to a relation of the form

$$\Delta t \Delta x \geq L^2, \quad L = \text{fundamental length}$$

.

Space-time UR

- Based on these arguments, Doplicher, Fredenhagen and Roberts (1994) arrived at uncertainty relations between the coordinates, which they showed could be deduced from a commutation relation of the type

$$[q_\mu, q_\nu] = iQ_{\mu\nu}$$

  where q_μ are self-adjoint coordinate operators, μ, ν run over space-time coordinates and $Q_{\mu\nu}$ is an antisymmetric tensor, with the simplest possibility that it commutes with the coordinate operators.
- We shall consider noncommutative geometry as a model of a quantum space-time.
- Our focus will be on QNM's in the NC framework.

Moyal

An example of noncommutative geometry is provided by the d-dimensional Groenewold-Moyal spacetime or GM plane, which is an algebra $\mathcal{A}_\theta(\mathbb{R}^N)$ generated by elements $\hat{x}_\mu$ ($\mu \in [0, 1, 2, \cdots, N-1]$) with the commutation relation

$$[\hat{x}_\mu\,, \hat{x}_\nu] = i\theta_{\mu\nu}\mathbf{1}\,,$$

$\theta_{\mu\nu}$ being real, constant and antisymmetric in its indices. This algebra can be represented by functions of commuting variables with a twisted product

$$\mu_\theta(f \otimes g) = f * g = f\, e^{i/2 \overleftarrow{\partial}_\mu \theta^{\mu\nu} \overrightarrow{\partial}_\nu} g = \mu_0(F_\theta f \otimes g).$$

The $*$ product defines the associative but noncommutative algebra $\mathcal{A}_\theta(\mathbb{R}^N)$. The twist element is denoted by F_θ.

The metric for an asymptotically flat, spherically symmetric and static black hole in 3+1 dimensions is given by

$$ds^2 = -f(r)dt^2 + f^{-1}(r)dr^2 + r^2 d\Omega^2 = g_{s\,ij} dx^i dx^j$$

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where $R_{\mu\nu}$ is the Ricci tensor that depends on the metric $g = g_s + h$ and g_s is the Schwarzschild background metric, while h is the metric perturbation.

QNM

In order to find a NC analog of the linearized vacuum Einsten equation, we use the NC differential geometry framework and choosed a twist given by

$$F_q = 1 \otimes 1 - \frac{iq}{2}\partial_r \wedge \partial_\phi + \mathcal{O}(q^2), \quad q \in \mathbb{R}$$

where q is the NC deformation parameter. With this, the wave equation for the QNM reduces to

$$\frac{d^2\psi}{dr_*^2} + \Big(\omega^2 - V(r)\Big)\psi = 0$$

where $V(r) = V_{RW} + V_q$ is the *quantum* Regge-Wheeler potential given by

$$V(r) = \frac{(r-R)\big(\ell(\ell+1)r - 3R\big)}{r^4} + qm\frac{\ell(\ell+1)(3R-2r)r + R(5r-8R)}{2r^5}.$$

QNM

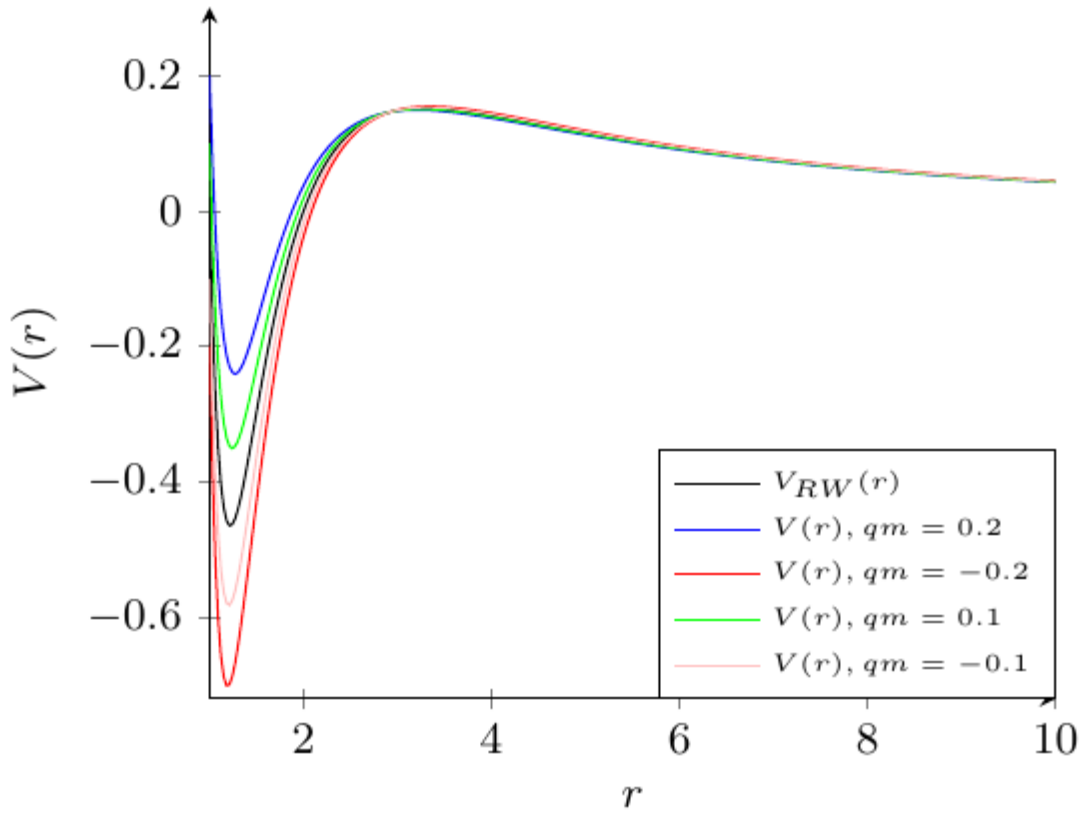


FIG. 1: The figure is drawn for $R = 2$ $(M = 1)$ and $\ell = 2$. We see that the behavior is very similar to the commutative case outside the horizon, and especially around the peak $r_0 \cong 1.5R$ and for larger r.

QNM

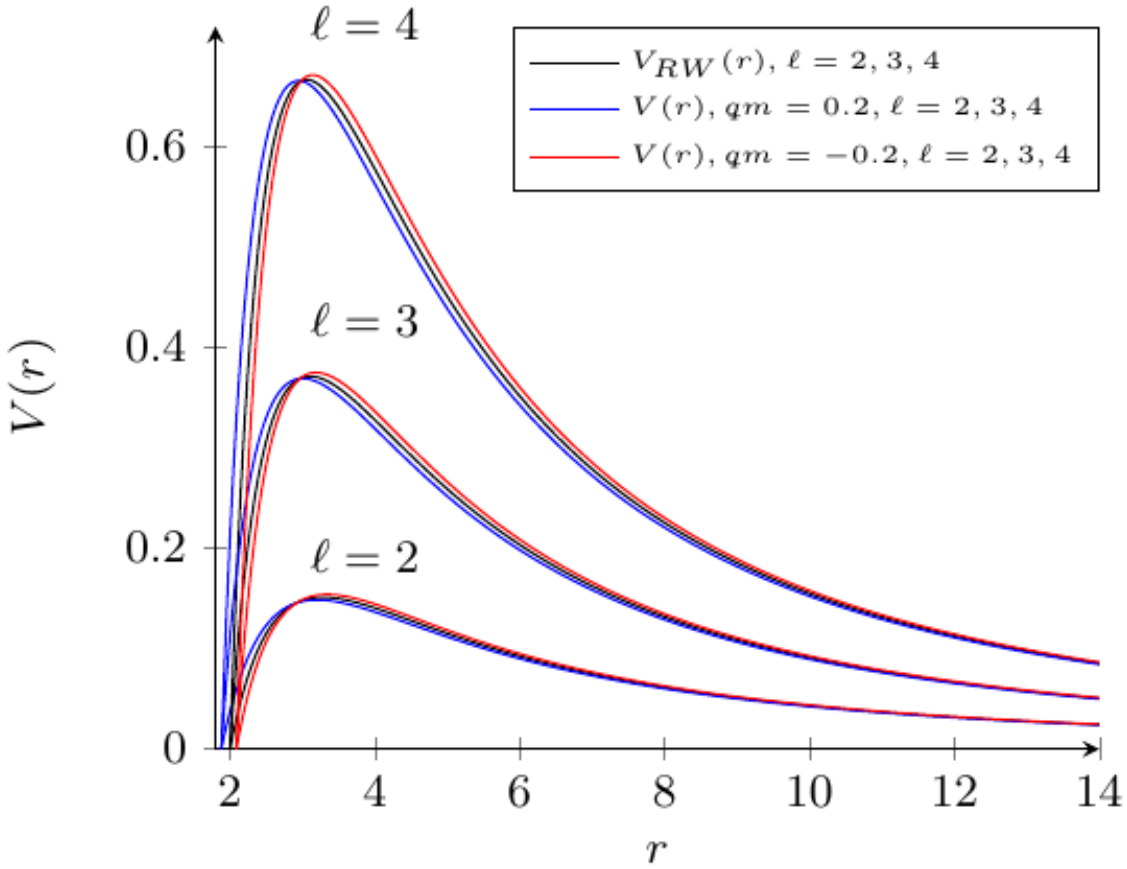


FIG. 2: The figure is drawn for $R = 2$ $(M = 1)$ and $\ell = 2, 3, 4$ around the peak.

QNM

$qm =$	-0.2	-0.1	0	0.1	0.2
$\ell = 2$					
$\mathrm{Re}(\omega_0)$	0.37712	0.37507	0.37316	0.37144	0.36991
$-\mathrm{Im}(\omega_0)$	0.08873	0.08906	0.08922	0.08898	0.08782
$\ell = 3$					
$\mathrm{Re}(\omega_0)$	0.60260	0.60076	0.59927	0.59827	0.59800
$-\mathrm{Im}(\omega_0)$	0.09243	0.09265	0.09273	0.09246	0.09132
$\ell = 4$					
$\mathrm{Re}(\omega_0)$	0.81218	0.81038	0.80910	0.80852	0.80897
$-\mathrm{Im}(\omega_0)$	0.09390	0.09410	0.09417	0.09397	0.09310

FIG. 3: The table of QNM's for $n = 0$, $M = 1$, $\ell = 2, 3, 4$.

QNM-Summary

- It is possible to calculate the NC corrections to QNM of Schwarzschild black hole in a mathematically consistent framework.
- QNM calculations for the more realistic Kerr black hole in the NC setup is under investigation.
- Logically q should be very small compared to any scales that have been observed so far. So can there ever be any possibility of detecting the effects of quantum spacetime?
- Around 1975 when the idea of GW detection was formally conceptualized, the accuracy in distance measurement was 1 part in 10^6, whereas presently in LIGO it is 1 part in 10^{18}.
- It is thus not totally inconcievable that future GW experiments would be able to detect (or rule out) physical effects of NC (or other quantum analogues) on space-time and gravity.

Other observations

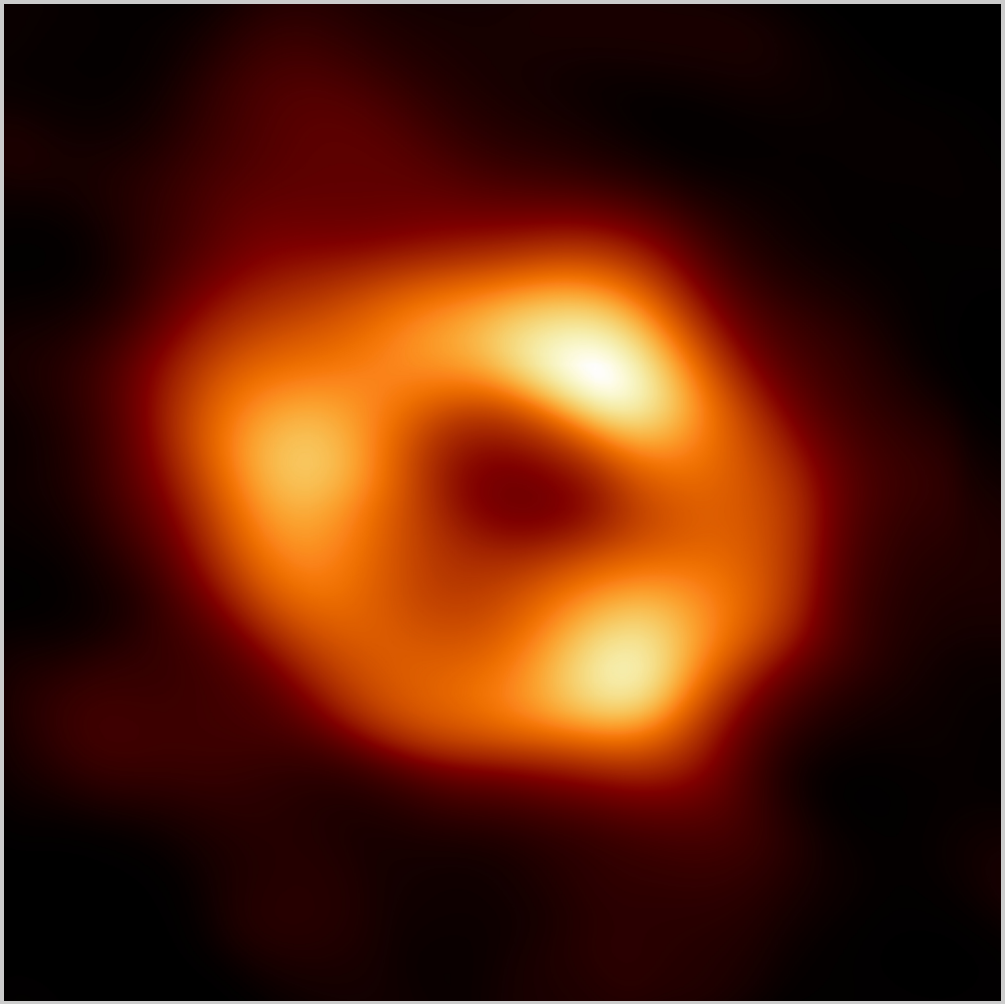
- LIGO is sensitive to data in the range of 10-10000 Hz with peak sensitivity around several 100 HZ.
- Other observatories such as NONOGRAV, based on pulsar timing arrays are sensitive to GW in the nanoHz range.
- Such low frequency and long wavelegth GW are expected from the merger of supermassive black holes with mass equivalent of 10^6 to 10^{10} solar masses.
- They are believed to exist at the galactic centers and in 2020 Reinhard Genzel and Andrea Ghez received the Nobel prize for analyzing a supermassive black hole a region called Sagittarius A^* at the centre of our galaxy Milky Way.

EHT 2019



SMBH in Messier 87 Galaxy, $6.5x10^9$ solar mass (EHT 2019).

EHT 2022



Sagittarius A^* at the center of Milky Way, $4x10^6$ solar mass (EHT 2022).

Nanograv 2023

- In 2023 NONOGRAV published a 15 year data set which shows the evidence for a low frequency GW background they claim is consistent with a population of supermassive black hole binaries distributed over the universe (arXiv:2306.16220 [astro-ph.HE]).
- They are all consistent with massive spinning Kerr black holes as predicted by GR.
- Interestingly, these supermassive black holes have also been found in very early universe.

JWT 2025 - early SMBH



Researchers using the NASA/ESA/CSA James Webb Space Telescope have confirmed an actively growing supermassive black hole within a galaxy just 570 million years after the Big Bang. Part of a class of small, very distant galaxies that have mystified astronomers, CANUCS-LRD-z8.6 represents a vital piece of this puzzle that challenges existing theories about the formation of galaxies and black holes in the early Universe. Ref: esawebb.org

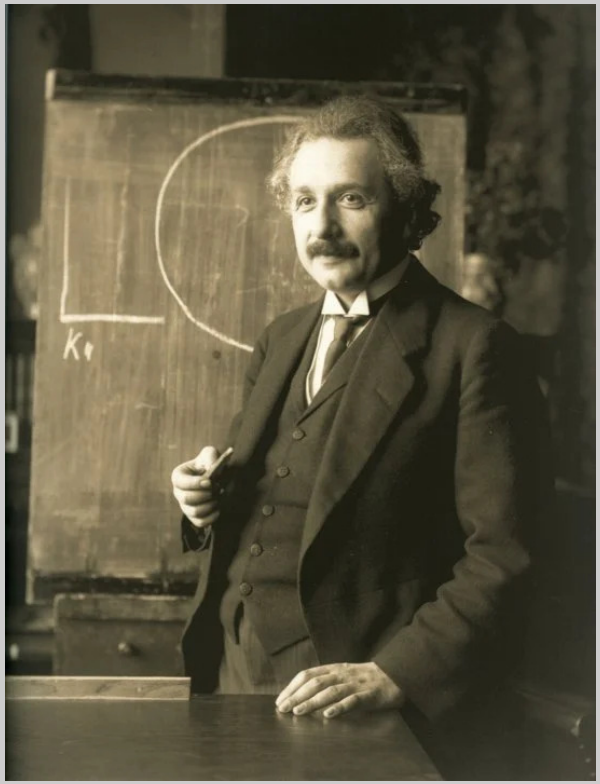
PBH seeding SMBH - NC effects

- How could such supermassive black holes have formed so early in the universe, whose age is about 13.8 billion years.
- There are ideas that primordial black holes in the very early universe could merged successively and might have acted as seeds of these supermassive black holes , provided the PBH can remain stable upto a suitable time.
- Stability of PBH has been studied from many angles, including using GUP inspired by noncommutativity and string theory.
- In our approach we study the stability of the PBH under gravitational perturbation in the presence of NC.
- Preliminary results indicate that the QNM's of a PBH can be less strongly damped and hence longer lived in presence of the NC effects.
- Such primordial black holes might decay more slowly in quantum space-time thereby allowing the seeding of SMBH, which is presently under investigation.

Outlook

- From the formulation of GR in 1916, we have come a long way in experimentally validating its predictions.
- Multimessanger astronomy is vigorously opening up new vistas in our understanding of the universe.
- Discovery of black holes and gravitational waves are at the forefront of such experimental advances.
- There have also been many ideas of the quantum nature of space-time, expected to be relevant for very early universe.
- Gravitational waves from early universe and from the primordial black holes might shed light on physical features of quantum spacetime, possibly through future GW detectors.
- Ongoing research in these areas is a part of the continuous exploration of the mysteries of the universe.

Einstein



The eternal mystery of the world is its comprehensibility.
The fact that it is comprehensible is a miracle.

A. Einstein, "Physics and Reality" Journal of the Franklin Institute,1936